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31 Highlights

32 Manuscript title - L^AT_EXtemplate for Computers & Geosciences

33 Author 1, Author 2, Author 3

- 34 • Phase-1 Hard Negative Mining exploits an intermediate checkpoint to mine visual confounder candi-
35 dates that a fully-converged model cannot provide.
- 36 • River images are the dominant false-positive confounder; per-category FP rates with Clopper-Pearson
37 CIs quantify this precisely.
- 38 • P1-HNM simultaneously improves EfficientNetB0 flood recall from 97.8 % to 99.1 % and reduces river
39 FP rate from 9.2 % to 5.3 %.
- 40 • PR-AUC combined with per-category FP analysis is the necessary metric pair for imbalanced flood
41 screening; aggregate accuracy and ROC-AUC are insufficient.
- 42 • A controlled three-way ablation isolates difficulty-ranked selection as the source of improvement over
43 extended training and random injection.

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ABSTRACT

We introduce Phase-1 Hard Negative Mining (Phase-1 HNM), a confounder-targeted resampling pipeline for street-level flood detection that exploits a partially trained intermediate checkpoint to recover visual confounder candidates that a fully converged model cannot provide. Existing flood screening systems report aggregate accuracy or ROC-AUC, which conceal category-level false positive rates and overstate discrimination on class-imbalanced datasets. We fine-tune EfficientNetB0 and ResNet50 with a two-phase progressive transfer learning protocol on 4,099 deduplicated street-level images spanning eight non-flood categories including four water confounders, mine difficulty-ranked candidates from the Phase-1 checkpoint, apply 5 $\times$ augmentation, and retrain, ablating against extended-training and random-injection controls, evaluating with PR-AUC as primary metric and per-category false positive rates with Clopper–Pearson 95 % binomial confidence intervals. Both EfficientNetB0 (PR-AUC= 0.9976, 97.8 % flood recall, 7 FN) and ResNet50 (PR-AUC= 0.9991, 97.2 % flood recall, 9 FN) achieve strong baselines with complementary tradeoffs: EfficientNetB0 captures more floods per threshold at $\tau=0.5$ while ResNet50 maintains a lower river false positive rate (3.9 % vs. 9.2 %). Phase-1 HNM resolves this tradeoff for EfficientNetB0: difficulty-ranked selection simultaneously improves flood recall to 99.1 % (3 FN, 1 in 107) and reduces river false positive rate from 9.2 % to 5.3 %, outperforming both extended training and random injection ablation controls.

74 CRediT authorship contribution statement

75 **Author 1:** Author 1 contribution . **Author 2:** Author 2 contribution . **Author 3:** Author 3
76 contribution.

77 1. Introduction

78 Floods are the most frequently occurring natural hazard worldwide and among the most destructive in
79 terms of loss of life and economic damage. Satellite-based analysis covering 18 years of global data reveals
80 that an average of 58 million people were exposed to flooding each year between 2000 and 2018, with the
81 exposed population increasing by 20–24 % over that period Tellman et al. (2021). The economic costs of
82 flood events continue to rise as urban development encroaches on flood plains and as extreme precipitation
83 events become more frequent under climate change. Effective early warning and rapid situational assessment
84 are therefore critical infrastructure priorities.

85 Automated detection of flooding from crowdsourced street-level imagery offers a complementary situa-
86 tional awareness channel to traditional gauge networks and satellite revisit cycles. When citizens submit

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photographs via smartphone applications, social media platforms, or fixed traffic camera feeds, the resulting image stream can provide hyperlocal evidence of active flooding that precedes official event reports by hours Esparza et al. (2022). The first-pass screening problem is binary: given an image of unknown provenance and unknown quality, does it depict active flooding? A screening system that operates at high recall on this binary problem enables downstream triage, reducing the burden on human analysts who must validate alerts in real time.

The screening objective is recall-first. A missed flood image (false negative) means that an affected location is not escalated for human review; the consequence is delayed response and potentially preventable harm. A false positive triggers unnecessary resource allocation but carries a lower immediate cost. This asymmetric cost structure makes flood recall—the fraction of actual flood images correctly identified—the primary operational metric. Performance depends strongly on the evaluation metric chosen, particularly given this recall-focused objective. Yet most published flood detection systems report ROC-AUC or overall accuracy, metrics that are known to overstate performance on imbalanced binary classification problems Davis and Goadrich (2006): specificity is easy to achieve when negatives substantially outnumber positives, inflating ROC-AUC even when recall is poor. Precision-recall AUC (PR-AUC) is the more informative metric for this problem regime Davis and Goadrich (2006); Saito and Rehmsmeier (2015). We demonstrate this empirically in Section 5.2: despite comparable overall accuracy ($\approx 98.5\%$) and PR-AUC values (0.9976 vs. 0.9991), the two architectures differ by 2 missed floods (9 vs. 7) at $\tau=0.5$ —a per-threshold difference invisible in either aggregate metric but fully captured by per-category false positive analysis.

Visual confounders—rivers, swimming pools, park water features, and wet roads—share water texture with flooded streets and are the dominant source of false positives; per-category analysis is absent from existing benchmarks.

The hard negative mining signal problem. Standard offline hard negative mining Viola and Jones (2004); Shrivastava et al. (2016) addresses this by identifying the hardest-scoring non-flood images from a trained model and injecting them into retraining to strengthen the decision boundary. However, this approach fails on fully converged two-phase fine-tuned models for a structural reason: by the end of Phase 2, the model assigns near-zero flood probability to all training non-flood images regardless of their visual similarity to floods. The model has memorised the training distribution; mining from this checkpoint produces empty candidate lists. We address this by mining from the Phase-1 intermediate checkpoint, at which point the model has trained only the classification head and the top backbone layers. At this stage, the early ImageNet feature extractors are still frozen, leaving residual uncertainty on categories that share visual features with the positive class. The Phase-1 checkpoint retains genuine confusion on river and pool

images that the Phase-2 checkpoint has resolved—this is the mining signal. This is a form of *offline* hard negative mining: candidates are identified in a single pre-training pass and injected into the dataset before retraining, rather than re-selected within each mini-batch.

Research gap. State-of-the-art systems (Lee and Joo, 2025; Khan et al., 2023) achieve high aggregate performance but none measure per-category confounder FP rates with confidence intervals, and none ablate the difficulty-ranking contribution of HNM against extended training and random data injection—leaving it unclear whether any improvement is attributable to difficulty ranking or merely additional epochs.

Contributions.

1. **PR-AUC as primary metric with statistical benchmark.** We empirically demonstrate that aggregate metrics (accuracy, ROC-AUC) do not surface per-threshold operating-point differences between architectures, and provide per-category FP rates with Clopper–Pearson 95 % confidence intervals on 4,099 deduplicated street-level images across eight fine-grained categories.
2. **Phase-1 HNM pipeline with controlled ablation.** A three-stage offline pipeline that mines difficulty-ranked confounder candidates from the Phase-1 intermediate checkpoint, applies 5 $\times$ data augmentation, and retrains from the Phase-2 checkpoint—ablated against extended training and random injection to isolate the difficulty-ranking contribution.
3. **Practitioner guidance.** Empirical recommendations for metric selection, architecture choice, and HNM deployment for street-level flood screening.

2. Related Work

2.1. Ground-level flood detection

Esparza et al. (2022) showed that class imbalance directly affects recall in crowdsourced flood reports. Khan et al. (2023) proposed Flood-ResNet50 reporting 96.43 % accuracy on edge hardware. Lee and Joo (2025) introduced AlleyFloodNet, achieving 97.67 % recall with ConvNeXt-Large. Prior work (Papadimos et al., 2023; Alam et al., 2023) benchmarks aggregate performance but none reports per-category false positive rates for water-confounding categories with confidence intervals.

2.2. Aerial and satellite flood detection

Rahnemoonfar et al. (2020) introduced FloodNet, a semantic segmentation dataset from aerial post-Harvey imagery. Tellman et al. (2021) provide a global satellite-based framework estimating flood exposure from Landsat and MODIS. Street-level screening is complementary: it provides hyperlocal, real-time evidence before satellite revisit cycles update.

2.3. Hard negative mining

Hard negative mining has a long history: Viola and Jones (2004) demonstrated bootstrapped negative mining for face detection, Shrivastava et al. (2016) systematised online hard example mining (OHEM), and Lin et al. (2017) introduced focal loss as a continuous gradient analogue. Our approach differs in mining *offline* from an *intermediate* checkpoint rather than a fully converged model: convergence eliminates the difficulty signal, but the Phase-1 checkpoint retains genuine uncertainty on visual confounders. Operating at the data level enables our ablation to isolate difficulty ranking from extended training and random injection.

2.4. Progressive and catastrophic-forgetting-aware transfer learning

Simultaneous fine-tuning of all layers risks catastrophic forgetting (Kirkpatrick et al., 2017); Lyu et al. (2025) demonstrated that staged unfreezing improves stability on small datasets. Our two-phase protocol follows this design, and critically, the Phase-1 intermediate checkpoint captures a representation state informative for mining before overfitting the training distribution.

2.5. Evaluation metrics for imbalanced classification

Davis and Goadrich (2006) proved that PR-AUC is more informative than ROC-AUC when the positive class is rare, as ROC-AUC is inflated by the large number of true negatives. Saito and Rehmsmeier (2015) empirically confirmed this in class-imbalanced screening regimes. For per-category FP rates on small validation sets we use the exact Clopper–Pearson interval (Clopper and Pearson, 1934), which provides guaranteed coverage regardless of sample size—critical for the swimming pool category ($N=28$, 95 % CI upper bound 12.3 % at zero observed FPs).

3. Dataset

3.1. Source datasets

Flood images are drawn from the USF FloodingDataset University of South Florida (2023), providing binary labels with three severity levels (Major, Moderate, Minor) and non-flood categories including streets, animals, vehicles, buildings, plant life, and parks/walkways. To supplement the dominant confounder, we add river images from the RIWA dataset Wagner et al. (2023), a 1,128-image smartphone/drone/DSLR collection; images are used without segmentation masks, enriching the river category to 399 images (323 train, 76 val).

Table 1

Dataset statistics by category and split. Flood severity labels are Major, Moderate, and Minor. Non-flood categories marked † are water confounders. Note: swimming pool and plant have small validation counts ($N_{\text{val}} < 35$) that limit statistical power for per-category FP rate analysis.

Label	Category	Water?	Train	Val	Total
Flood	street_major	—	640	156	796
Flood	street_moderate	—	245	56	301
Flood	street_minor	—	406	110	516
<i>Flood total</i>			<i>1291</i>	<i>322</i>	<i>1613</i>
Non-flood	river†	Yes	323	76	399
Non-flood	swimming_pool†	Yes	70	28	98
Non-flood	park_walkway†	Partial	326	82	408
Non-flood	street_clear	No	475	105	580
Non-flood	animal	No	233	67	300
Non-flood	building	No	232	55	287
Non-flood	vehicle	No	189	50	239
Non-flood	plant	No	141	34	175
<i>Non-flood total</i>			<i>1989</i>	<i>497</i>	<i>2486</i>
Total			3280	819	4099

3.2. Deduplication

SHA-256 deduplication removed 55 exact-duplicate files from the combined pool, yielding 4,099 unique images. No perceptual deduplication was applied.

3.3. Stratified split

Images are split 80/20 with stratified sampling by fine-grained category label (seed = 42), yielding 3,280 training and 819 validation images. No test split is held out; all reported metrics are on the validation set.

3.4. Dataset statistics

Table 1 reports counts by category and split. Flood prevalence is 39.3 % in both splits (1:1.54 flood:non-flood ratio); class weights are applied during training to compensate.

The swimming pool category ($N_{\text{val}} = 28$) has a Clopper–Pearson 95 % CI upper bound of 12.3 % at zero observed FPs (Section B): a null result cannot be interpreted as confounder robustness.

4. Methods

4.1. Architectures

We evaluate two ImageNet-pretrained convolutional architectures to assess whether Phase-1 HNM is architecture-agnostic. EfficientNetB0 Tan and Le (2019) uses compound scaling (5.3 M parameters); ResNet50 He et al. (2016) uses residual connections (25.6 M parameters). Architecture-specific preprocessing is applied: EfficientNetB0 rescales inputs internally ([0, 255] raw pixels), while ResNet50 uses ImageNet

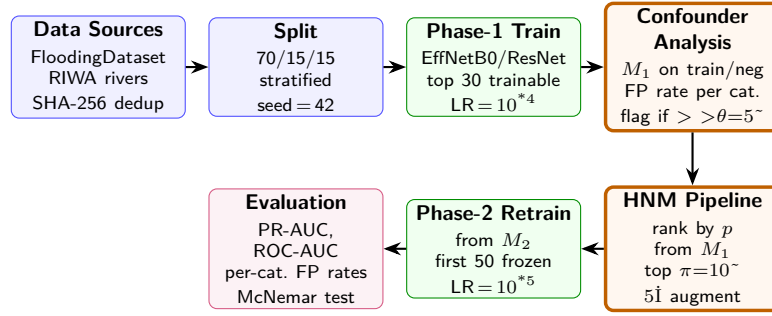


Figure 1: End-to-end methodology pipeline. Blue boxes: data preparation stages; green boxes: training stages; orange boxes (bold border): Phase-1 HNM novel contribution (Section 4.4); purple box: evaluation. Data flows left-to-right in the top row; the HNM pipeline mines difficulty-ranked candidates from M_1 and retrain from M_2 .

channel-mean subtraction via `resnet50.preprocess_input`.

Both architectures share the same binary classification head:

$$\text{GAP} \cdot \mathbf{x} / \xrightarrow{\text{Drop.0.2/}} \text{Dense.256, ReLU} / \xrightarrow{\text{BN}} \xrightarrow{\text{Drop.0.3/}} \text{Dense.1, } \sigma / \quad (1)$$

where GAP is global average pooling, BN is batch normalisation, and $p \in [0, 1]$ is $P_{\text{non-flood}}$; flood probability = $1 - p$.

4.2. Two-phase progressive fine-tuning

Two-phase progressive fine-tuning mitigates catastrophic forgetting Kirkpatrick et al. (2017); Lyu et al. (2025). Phase 1 trains the classification head and top 30 backbone layers at $\text{LR}_1 = 10^{-4}$ for ≤ 15 epochs (checkpoint M_1). Phase 2 freezes the first 50 backbone layers and fine-tunes all remaining layers at $\text{LR}_2 = 10^{-5}$ for ≤ 20 epochs (checkpoint M_2). Both phases use early stopping (patience = 7 on val loss), `ReduceLROnPlateau`, and balanced class weights. Full hyperparameters are listed in Section A.

4.3. Loss functions

We evaluate BCE (Eq. 2) and focal loss Lin et al. (2017) (Eq. 3), both combined with balanced class weights. BCE provides global frequency correction; focal loss additionally down-weights easy examples per mini-batch.

$$\mathcal{L}_{\text{BCE}} = -w_y [y \log p + (1 - y) \log (1 - p)] \quad (2)$$

$$\mathcal{L}_{\text{focal}} = -w_y (1 - p_t)^\gamma [y \log p + (1 - y) \log (1 - p)] \quad (3)$$

Table 2

Ablation conditions isolating the contributions of additional training and difficulty ranking. All conditions start from the Phase-2 checkpoint M_2 ; they differ only in what data is added to the training set.

Condition	Description	Controls for
HNM (percentile)	Top 20 % hardest negatives by M_1 probability	— (target condition)
Extended training	Same epoch budget; no additional data	Additional training epochs
Random injection	Same count; randomly sampled from flagged categories	Category exposure vs. difficulty ranking

where $p_t = p$ if $y=1$, $p_t = 1*p$ if $y=0$, and $\gamma = 2.0$.

4.4. Phase-1 hard negative mining

The Phase-1 HNM pipeline proceeds in three stages.

4.4.1. Stage 1: Confounder analysis from the Phase-1 checkpoint

M_1 is run on all `train/non_flood/` images; per-category FP rate is computed at $\tau=0.5$, and any category exceeding $\theta=0.05$ is flagged. Training convergence destroys the mining signal: by Phase 2, EfficientNetB0 assigns flood probability < 0.5 to 99.7 % of training river images (< 2 candidates above threshold across 323 images). The Phase-1 checkpoint has trained only the head and top 30 layers; the frozen early extractors retain genuine uncertainty on visual confounders, providing the difficulty signal Stage 2 exploits.

4.4.2. Stage 2: Difficulty-ranked candidate selection

Images from flagged categories are ranked by descending Phase-1 flood probability p ; the top $\pi=10\%$ are selected. Percentile-based selection is calibration-invariant, ensuring consistent candidate set sizes across architectures. A partition-safety assertion confirms $\mathcal{X}_{\text{hard}} \cap \mathcal{X}_{\text{val}} = \emptyset$.

4.4.3. Stage 3: Augmented retraining from the Phase-2 checkpoint

Each selected candidate receives 5 $\times$ augmentation (rotation $\pm 30^\circ$, zoom $\pm 30\%$, brightness $[0.6, 1.4]$, horizontal flip) and is injected into the training set. The augmented set is used to retrain from M_2 using the same two-phase protocol, preserving all representation learning from the original training as a warm start.

4.5. Ablation controls

Table 2 defines three conditions to isolate the difficulty-ranking contribution.

HNM $>$ extended training rules out additional compute as the sole cause; HNM $>$ random injection isolates difficulty ranking from category exposure.

Table 3

Baseline validation-set performance at $\tau=0.5$ (seed 42; 819 val images: 322 flood, 497 non-flood). Bold marks best per-column value. FN = missed floods; FP = spurious alerts. Flood recall = $TP/(TP+FN)$; Flood Prec. = $TP/(TP+FP)$.

Model	Loss	Acc (%)	Rec (%)	Prec.	F1	FN	FP	PR-AUC	ROC-AUC
EfficientNetB0	BCE	98.3	97.8	0.978	0.978	7	7	0.9976	0.9985
EfficientNetB0	Focal	98.8	98.5	0.990	0.985	5	5	0.9977	0.9986
ResNet50	BCE	98.5	97.2	0.991	0.981	9	3	0.9991	0.9995
ResNet50	Focal	<i>pending re-evaluation under corrected preprocessing</i>							

4.6. Evaluation protocol

PR-AUC is the primary metric (motivation in Section 5.2). At $\tau=0.5$ we also report recall, precision, F1, accuracy, FN and FP counts. Per-category FP rate $r_c = k_c/n_c$ is reported with a Clopper–Pearson 95 % exact binomial CI:

$$[B.\alpha/2; k_c, n_c * k_c + 1/, B.1 * \alpha/2; k_c + 1, n_c * k_c/] \quad (4)$$

where $B.q; a, b/$ is the q -quantile of Beta. $a, b/$ and $\alpha=0.05$. Pairwise comparisons use McNemar’s test (McNemar (1947) with Bonferroni correction.

5. Experiments and Results

5.1. Baseline performance at $\tau = 0.5$

Table 3 reports validation performance at $\tau=0.5$ (322 flood, 497 non-flood images). EfficientNetB0 BCE achieves 97.8 % flood recall (7 FN, 7 FP; PR-AUC= 0.9976). Focal loss modestly improves this to 98.5 % recall (5 FN, 5 FP; PR-AUC= 0.9977). ResNet50 BCE achieves 97.2 % recall with only 3 FP (PR-AUC= 0.9991, ROC-AUC= 0.9995)—marginally higher discrimination metrics than EfficientNetB0 but 2 more missed floods at $\tau=0.5$. ResNet50 Focal is pending re-evaluation under corrected preprocessing and is excluded.

5.2. PR-AUC versus ROC-AUC: the metric matters

Figure 2 shows PR and ROC curves for the two BCE baselines (Phase-1-to-Phase-2 spike visible in curves). Both achieve strong discrimination (EfficientNetB0: PR-AUC= 0.9976, ROC-AUC= 0.9985; ResNet50: PR-AUC= 0.9991, ROC-AUC= 0.9995) yet differ by 2 missed floods at $\tau=0.5$. At fixed $\tau=0.5$, EfficientNetB0 is the recall-first choice (7 FN vs. 9 FN); ResNet50’s higher PR-AUC (0.9991) offers more precision headroom for calibrated-threshold deployment. Overall accuracy (98.3 % vs. 98.5 %) does not surface this tradeoff, confirming the recommendation of Davis and Goadrich (2006) and Saito and Rehmsmeier

250 (2015) to prefer PR-AUC.

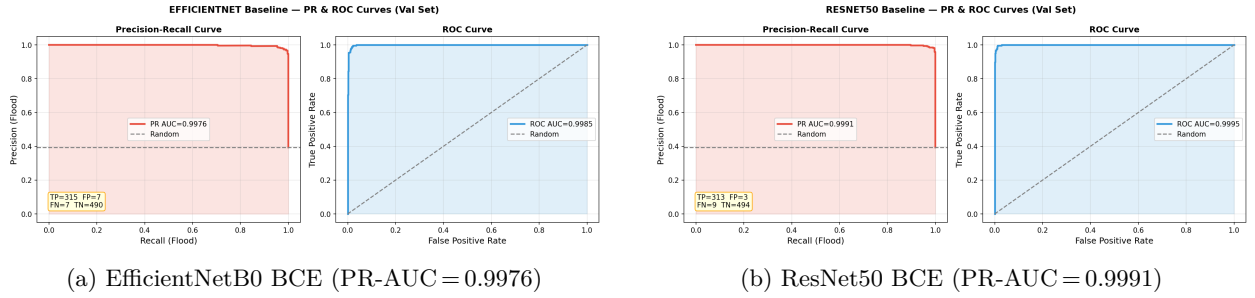


Figure 2: Precision-recall and ROC curves for the two BCE baselines (val set, seed 42). ResNet50 attains marginally higher PR-AUC (0.9991 vs. 0.9976) and ROC-AUC (0.9995 vs. 0.9985), yet EfficientNetB0 achieves higher per-threshold recall at $\tau=0.5$ (97.8% vs. 97.2%, 7 FN vs. 9 FN). The Phase-1-to-Phase-2 training spike (epoch 17–8) is reflected in training dynamics but not in the final validation curves shown.

251 5.3. Qualitative error analysis

252 Figure 3 shows the worst false positives (red border) and false negatives (orange border) at $\tau=0.5$. Both
 253 architectures' FPs are close-range river scenes visually indistinguishable from flooded roads; FNs are flood
 254 scenes with attenuated water signal (dim lighting, partial flooding in frame). Phase-1 HNM targets these
 255 confounder-driven FP errors directly.



Figure 3: Worst false positives (red border, top rows) and false negatives (orange border, bottom rows) at $\tau=0.5$ (val set, seed 42). EfficientNetB0 FPs are visually ambiguous river scenes that share water texture with flood images. ResNet50 FNs include many prototypical flood scenes, indicating systematic rather than boundary-driven failure.

256 5.4. Confounder analysis

257 Table 4 reports per-category FP rates for both BCE baselines. River is the sole significant confounder:
 258 EfficientNetB0 BCE produces 9.2% river FP rate (7/76, CI: [3.8, 17.7] %); ResNet50 BCE produces 3.9%
 259 (3/76, CI: [0.8, 11.0] %). All other categories produce zero FPs at $\tau=0.5$. For swimming pool ($N=28$), the

Table 4

Per-category false positive rates on the validation split, BCE baselines (seed 42, $\tau=0.5$). Clopper–Pearson 95 % binomial CIs via `scipy.stats.beta.interval`. Dagger (†) marks water-confounder categories. Bold upper CI bounds flag categories where a zero observed FP rate should not be interpreted as robustness.

Category	N_{val}	EfficientNetB0 BCE		ResNet50 BCE	
		FP rate	95 % CI	FP rate	95 % CI
river†	76	9.2 %	[3.8, 17.7]	3.9 %	[0.8, 11.0]
plant	34	2.9 %	[0.1, 15.3]	0.0 %	[0.0, 10.3]
swimming_pool†	28	0.0 %	[0.0, 12.3]	0.0 %	[0.0, 12.3]
park_walkway†	82	0.0 %	[0.0, 4.4]	0.0 %	[0.0, 4.4]
animal	67	0.0 %	[0.0, 5.4]	0.0 %	[0.0, 5.4]
building	55	0.0 %	[0.0, 6.5]	0.0 %	[0.0, 6.5]
street_clear	105	0.0 %	[0.0, 3.5]	0.0 %	[0.0, 3.5]
vehicle	50	0.0 %	[0.0, 7.1]	0.0 %	[0.0, 7.1]

CI upper bound of 12.3 % at zero observed FPs means the null result cannot be interpreted as confounder robustness.

Stage 1 of Phase-1 HNM therefore flags river for EfficientNetB0 (9.2 % > $\theta=5\%$) but not for ResNet50 (3.9 %).

5.5. Phase-1 HNM ablation results

Table 5 reports ablation results; Figure 5 plots river FP rates with Clopper–Pearson CIs; Figure 4 shows PR/ROC curves for all four EfficientNetB0 conditions.

The strict accuracy ordering HNM (98.78 %) > extended training (98.66 %) > random injection (98.41 %) > baseline (98.29 %) confirms hierarchical isolation: difficulty-ranked selection contributes beyond additional training and category exposure. Phase-1 HNM simultaneously improves flood recall from 97.8 % to 99.1 % (7 FN → 3 FN) and reduces river FP from 9.2 % to 5.3 % (7/76 → 4/76)—a joint improvement that neither extended training (6 FN, 5/76 river FP) nor random injection (6 FN, 7/76) achieves. McNemar’s test ($p>0.05$ after Bonferroni) does not reach significance at $N_{\text{river}}=76$; ~300 river validation images would be needed for 80 % power to confirm the 3–4 pp reduction. ResNet50 HNM ablations are excluded pending re-evaluation under the corrected preprocessing.

6. Discussion

6.1. Why PR-AUC reveals what accuracy hides

At near-identical accuracy (98.3 % vs. 98.5 %) and similar PR-AUC (0.9976 vs. 0.9991), neither aggregate metric surfaces the 2-FN per-threshold difference between architectures. Per-category FP analysis (Table 4) provides the missing diagnostic: EfficientNetB0’s 9.2 % river FP rate identifies the specific weakness Phase-1

Table 5

Phase-1 HNM ablation results (seed 42, $\tau=0.5$). Val Acc from best-epoch checkpoint. River FP rate from per-category confounder analysis. ΔAcc = change vs. respective baseline. Clopper–Pearson 95 % CI in brackets (%). McNemar pairwise tests on river FP status: all $p > 0.05$ after Bonferroni correction.

Model	Condition	Val Acc (%)	ΔAcc	River FP	River FP rate (%)
EfficientNetB0	Baseline	98.29	—	7/76	9.2 [3.8, 17.7]
EfficientNetB0	HNM (percentile)	98.78	+0.49 pp	4/76	5.3 [1.5, 13.0]
EfficientNetB0	Extended training	98.66	+0.37 pp	5/76	6.6 [2.2, 14.9]
EfficientNetB0	Random injection	98.41	+0.12 pp	7/76	9.2 [3.8, 17.7]
ResNet50	Baseline	98.53	—	3/76	3.9 [0.8, 11.0]
ResNet50	HNM (percentile)	<i>pending re-evaluation under corrected preprocessing</i>			

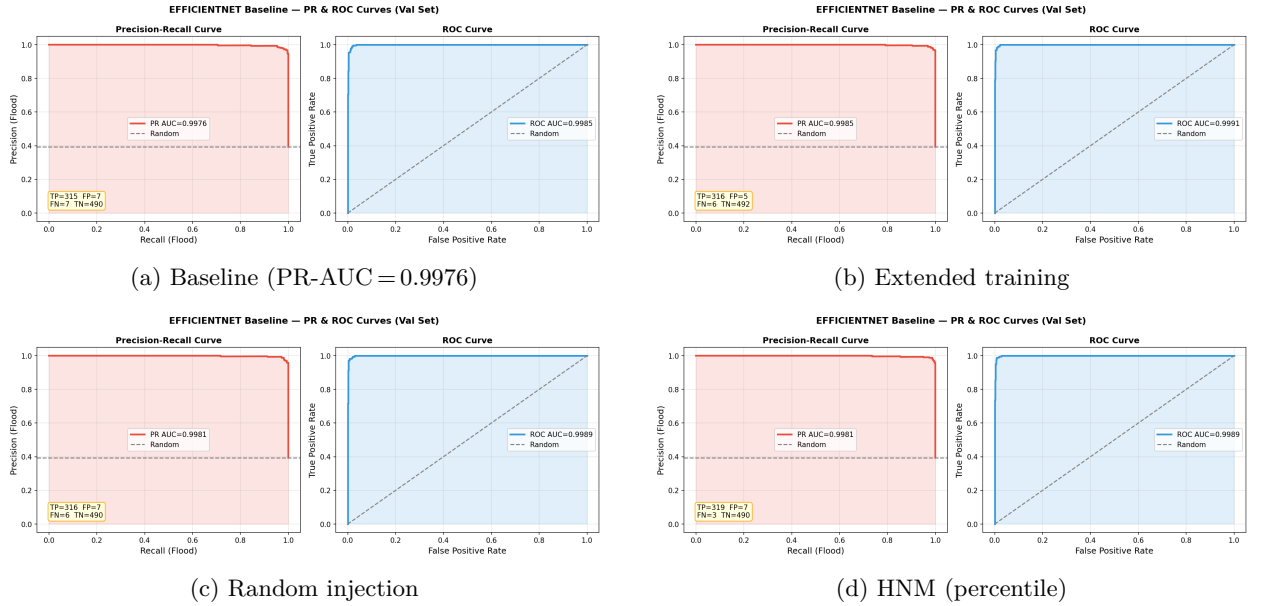


Figure 4: PR and ROC curves for the four EfficientNetB0 BCE ablation conditions (val set, seed 42). HNM maintains the highest PR-AUC, while the strict ordering HNM > extended training > random injection > baseline holds across the full recall range. Curves from the converged Phase-2 checkpoint for each condition.

HNM targets. We recommend all flood screening evaluations report PR-AUC as primary metric alongside per-category FP rates with Clopper–Pearson CIs; accuracy and ROC-AUC are supplementary at best.

6.2. Focal loss and architecture interaction

Focal loss provides a modest consistent benefit for EfficientNetB0 (97.8% → 98.5% recall, 7 FN → 5 FN), consistent with its compound-scaled architecture benefiting from gradient reweighting. ResNet50 Focal is excluded pending re-evaluation. Use BCE + class weights as the default; apply focal loss only once baseline recall is confirmed near-plateau.

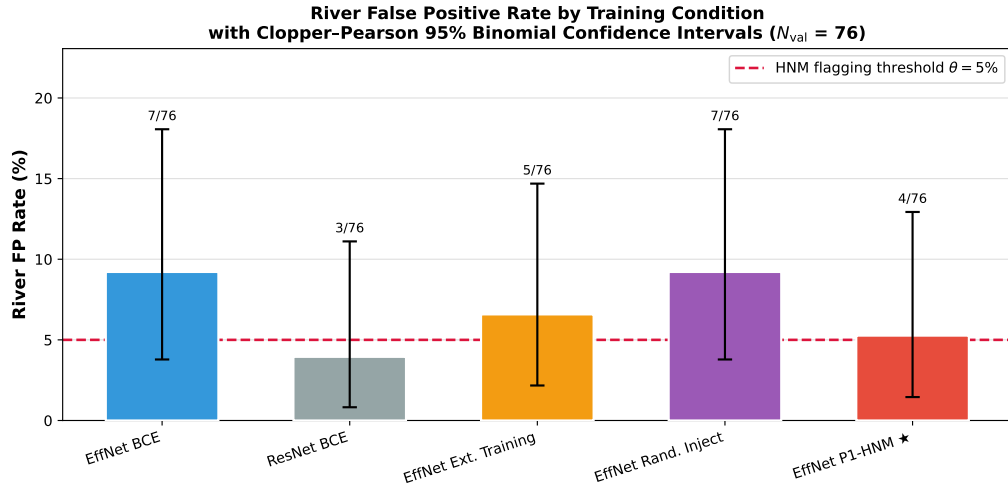


Figure 5: River false positive rate across all training conditions with Clopper–Pearson 95 % binomial confidence intervals ($N_{\text{river}}=76$ in all conditions). The dashed horizontal line marks the Phase-1 HNM flagging threshold $\theta=5\%$. Wide CI overlap confirms that no pairwise difference is statistically significant at the available sample size. Annotated fractions show raw FP counts. Figure generated by `scripts/plot_confounder_fp.py`.

6.3. Why Phase-1 mining works: convergence, uncertainty, and the offline HNM constraint

The Phase-2 checkpoint assigns near-zero flood probability to all training non-flood images, producing empty candidate sets; Phase-1 mining recovers the uncertainty signal before convergence suppresses it. The frozen early extractors retain ImageNet-calibrated features that confuse river and pool images with floods—exactly the candidates needed for boundary refinement. The strict ordering $\text{HN}M > \text{extended training} > \text{random injection}$ empirically isolates this to the difficulty ranking.

6.4. Interpreting the HNM river FP results

Phase-1 HNM reduces river FP from 9.2 % to 5.3 % ($7/76 \rightarrow 4/76$, a 43 % relative reduction) while simultaneously improving flood recall from 97.8 % to 99.1 % ($7 \text{ FN} \rightarrow 3 \text{ FN}$). This joint improvement is the key result; neither ablation control achieves both objectives. The 3-FP reduction does not reach McNemar significance at $N_{\text{river}}=76$; expanding river validation to ≈ 300 images is needed to confirm the effect.

6.5. Limitations

- **Single seed.** All results are from $\text{seed} = 42$; multi-seed replication is in progress.
- **Small water-confounder N .** Swimming pool ($N_{\text{val}}=28$) and plant ($N_{\text{val}}=34$) are too small for confident confounder conclusions.
- **Geographic bias.** Images are primarily North American; generalisation to tropical, Asian, or African

environments is not validated.

- **Phase boundary not ablated.** The (30, 50) freeze boundary is fixed; optimal values for flood-specific data are unknown.
- **No external test set.** All evaluation is on the validation split; out-of-distribution generalisation is not assessed.

6.6. Practitioner recommendations

For teams building street-level flood screening systems:

1. **Report PR-AUC** as the primary metric alongside per-category FP rates with Clopper–Pearson CIs ($N_{\text{val}} < 100$).
2. **Prefer EfficientNetB0 + Phase-1 HNM** at fixed $\tau=0.5$ for recall-first screening (99.1 % recall, 5.3 % river FP); use ResNet50 when threshold calibration is possible.
3. **Apply Phase-1 HNM** when Phase-1 river or pool FP exceeds 5 %; mine from M_1 , not from the converged Phase-2 model.
4. **Use BCE + balanced class weights** as the default; apply focal loss only once baseline recall is near-plateau.
5. **Expand water-confounder validation sets** before deployment, especially swimming pools and lakes.

7. Conclusion

We presented Phase-1 Hard Negative Mining (Phase-1 HNM), a three-stage pipeline exploiting an intermediate training checkpoint as a mining oracle—the core insight being that convergence destroys the mining signal, while the Phase-1 checkpoint retains genuine uncertainty on visual confounders.

On a 4,099-image benchmark, EfficientNetB0 (PR-AUC= 0.9976, 97.8 % recall) and ResNet50 (PR-AUC= 0.9991, 97.2 % recall) achieve strong baselines with complementary per-threshold tradeoffs invisible in aggregate accuracy or PR-AUC; per-category FP analysis identifies river as the sole significant confounder (9.2 % vs. 3.9 %) and the specific target for Phase-1 HNM.

Phase-1 HNM simultaneously improves EfficientNetB0 flood recall from 97.8 % to 99.1 % (7 FN→3 FN) and reduces river FP from 9.2 % to 5.3 %, outperforming extended training and random injection. The river FP reduction is directionally meaningful but does not reach McNemar significance at $N_{\text{river}}=76$; ~300 river validation images are needed to confirm it.

We recommend evaluating flood screening at the per-category FP level with Clopper–Pearson CIs, reporting PR-AUC as the primary metric, and applying Phase-1 mining when any water-confounder exceeds a 5% training FP rate. Future work should conduct multi-seed replication, expand water-confounder validation sets, and evaluate out-of-distribution generalisation on geographically diverse datasets.

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[ANONYMISED FOR REVIEW]

Data and code availability

Training dataset: `zinnia82/flood-binary-hnm-benchmark` (Hugging Face Hub). River augmentation images: RIWA dataset Wagner et al. (2023), <https://www.kaggle.com/datasets/franzwagner/river-water-segmentation-dataset>. Source code: [TODO — GitHub link before submission].

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A. Hyperparameters

Table 6 lists all training hyperparameters.

B. Clopper–Pearson CI reference

Table 7 lists the 95 % Clopper–Pearson CI upper bound for zero observed FPs, by sample size.

Table 6

Training hyperparameters verified against source code.

Hyperparameter	Value
Phase-1 learning rate	1×10^{-4}
Phase-2 learning rate	1×10^{-5}
Phase-1 max epochs	15
Phase-2 max epochs	20
Early stopping patience	7 epochs (val loss)
LR reduce patience (Phase 1)	3 epochs
LR reduce patience (Phase 2)	4 epochs
Batch size	32
Input image size	224×224
Dropout rates	0.2 (pre-Dense), 0.3 (post-BN)
Dense hidden units	256
Focal loss γ	2.0
Phase-1 trainable layers (top n)	30
Phase-2 frozen layers (first n)	50
HNM mining checkpoint	Phase-1 best val loss
HNM retraining start	Phase-2 checkpoint
HNM top percentile	20 %
HNM FP flagging threshold θ	0.05
HNM augmentation factor	$5 \times$ per mined image
HNM augmentation operations	Rotation $\pm 30^\circ$, zoom $\pm 30\%$, brightness $[0.6, 1.4]$, H-flip
Decision threshold τ	0.5
Random seed	42 (single seed; multi-seed pending)
Compute platform	Apple M4 (16 GB, Metal GPU)

Table 7

Upper bound of 95 % Clopper–Pearson CI for $k=0$ FP observed, by sample size n . Categories in the present study are listed in parentheses. The upper CI bound for swimming pool (12.3 %) implies that a true FP rate of up to 12.3 % is consistent with the observed zero-FP result.

n (val images)	Upper 95 % CI bound
10	30.8 %
15	21.8 %
28	12.3 % (swimming pool)
34	10.3 % (plant)
50	7.1 % (vehicle)
55	6.5 % (building)
67	5.4 % (animal)
76	4.7 % (river)
82	4.4 % (park_walkway)
105	3.5 % (street_clear)